Metadata

Data Repository on Mendeley Data

Description of Dataset

• *"The dataset holds the raw data traces of the bouton responses presented in Bilz et al. We analyzed synaptic boutons of Kenyon cells of the Drosophila mushroom body γ-lobe, two to five by expressing a fluorescent Ca2+ sensor in single Kenyon cells so that axonal boutons could be assigned to distinct cells. Single files in this data set represent single cells in the described experiments. The data set consists of all cells included in the publication and is devided into a raw (.xlsx files) and analysed (MatLab files) dataset."*

• So far we use the raw .xlsx files

• *"Each file contains responses of one cell. The data inside the file is split onto two sheets: Sheet 1 contains pre learning phase data, Sheet 2 contains post learning data. The data on each sheet is saved as a two dimensional matrix mxn, where m represents the number of acquired frames and n the number of identified boutons on the cell. The first row of the sheet contains stimulus inforation."*

Data Collection

• KC Labeling and Calcium sensors:

◇ Fluorescent Ca^{2+} sensor sparsly expressed in sigel γ KC (MARCM)
  ■ GCaMP3 as cytosolic Ca^{2+} sensor of specific to gamma KCs
  ■ mCherry co-expressed in GCaMP3 labeled cells as synaptic calcium sensor and anatomical marker for synaptic boutons (See Normalisation)
◇ Expression was limited to 1–3 KCs per brain hemisphere per fly

• n KC and Butons

◇ Labeling was successful in ~50% of flies (120 out of 240 total)
  ■ n Labeld flys = 120
  ■ n labeld KCs = 270
  ■ n imaged KCs = 139
  ■ n responsive KCs = 49 (responsesd to at least one of the three tested odorants or the diluent)
  ■ n analysed KCs = 31   (KC axons containing 38–93 synaptic boutons (mean ±SD; 62.74 ± 12.61) that could be anatomically/functionally characterized)

  ■ Spesific response rates:
    - HCH = 8.1%   => 11 KCs
    - 3-OCT = 10.0%   => 14 KCs
    - 1-OCT = 4.1%   => 6 KCs
      → Responiive to at least one of the Ordants = 31 KCs
      → OIL = 2.2% => 3 Kcs
      → SCHOWN IN FIG 3, S1 and S3

• Aversive Olfactory conditioning:

◇ Definitions

- Flies learn to associate an odor (CS) with punishment (US).
- Bliz et al 2020 measures odor responses **before and after training in the same individual.**

- **CS^{+}** = CS + US
- **CS^{-}** = CS - US
- **Control condition** = Backward conditioning
  - US precedes CS (Shock before odor)
  - *Not neutral* induces learned **attraction** to the odor (relief leering)

◇ Odorants:

→ Mineral oil (OIL in .xls data)
→ 4-methylcyclohexanol (MCH in .xls data)
→ 3-octanol (3-OCT in .xls data)
→ 1-octanol (1-OCT)

◇ Conditioning Procedure

- n labeld Flys used = 18
- n responsive KCs to any ordant = 20
- n responsive KCs to MCH and 3-OCT = 4 => These were KCs used to create the Forward conditioning recoring data !!!
- This comprises the forwardCondtioning data

- **CS+ presentation**:
  - **Odor (MCH or 3-OCT)** presented 10 s after imaging start, for 60 s
  - **Electric shocks** (90 V, 1.25 s duration, 3.75 s interval) delivered during CS+:
  - **12 shocks total**, starting **5 s after CS+ onset**

- **CS^{-} presentation**:
  - **The other odor (MCH or 3-OCT, whichever was not CS+)**
  - Presented after a 60 s rest, also for 60 s
  - No shocks paired with CS^{-}
  - QUESTION: WHAT ODOR IS CS+/ CS- IN THE DATA ? (cant find a label)

- Imaging offset 30 s after odor offset

- **Control Group**
  - Same stimulation protocol (odor presentation, imaging timing), but without electric shocks

    ⇒ THIS IS AS I UNDERSTAND NOT BACKWARD CONDITIONING

    • That was used to assess **odor-evoked responses without associative learning** (see Figure 5A) TO DO recheck that

• Backwards Conditioning:

◇ Definitions:

- Repeated experiment with US (shock) preceding CS (odor) (backward conditioning)
- Uesed as controle procedure to forward condtioning

- Testing if temporal relation CS / US determines bouton activity decorrelation
- Different individuals used here (Animals die during recording!)

■ Procedure not neutral
- Relive learning, odor indicates the absence of punichment
→ Novel finding: CS conditions; more boutons showed an increase in activity in the g4 compartment -> TO DO = Plot this in data

◇ **Conditioning Procedure / N:**

◇ n Addtionaly tested flys for backwards condtioning = 293

◇ n responive KCs to any ordand = 12 (KCs shown in fig S6)

◇ n respnsive to MCH and 3_OCT = 4 => These are the recording data form the “backwardcondtionind” data

• Odor Response Recording:

◇ **Imaging Setup:**

■ g1 was not visualizable due to optical inaccessibility

■ Traces from a single g1 Kenyon cell (GCaMP3) and its boutons (mCherry) were recorded simultaneously

■ Imaging was performed at a frame rate of **4 Hz**. *(1 Frame = 0.25s)*

■ Only KCs showing robust responses > 3 std above the pre baseline, were considered responsive.

◇ **Odorants:**

→ Mineral oil (OIL in .xls data)
→ 4-methylcyclohexanol (MCH in .xls data)
→ 3-octanol (3-OCT in .xls data)
→ 1-octanol (1-OCT)

◇ **Odor Recording:**

■ One recording per odorant per focal plane
- Duration 21.25 s => 21.25 at 4hz => 85 Frames long recording

■ Odor onset at 6.25 s
- => 6.25 at 4hz => Onset at Frame 25

■ Odor presentation duration 2.5 s
- 2.5s at 4hz => 10 Frames stimulation
- stimulation offset at Frame 35

■ 20 s interval between each measurement

■ Imaging performed in 3–7 focal planes

◇ **Post-Training Imaging:**

■ After training, flies rested for 3 minutes

■ Odor-evoked Ca^{2+} transients measured again in all focal planes (3-7 individual planes), same as pre-training

• Normalization Procedure:

■ $\Delta F/F_0$ was calculated for each bouton per KC for both the GCaMP3 (cytosolic) and mCherry (synaptic) calcium signals.

- Signal correction:
  - The $\Delta F/F_0$ of mCherry (synaptic) was subtracted from the $\Delta F/F_0$ of GCaMP3 (cytosolicl) to isolate bouton-specific calcium dynamics.
    → Observation: GCaMP3 also detected odor-evoked Ca^{2+} influx between boutons, but with lower amplitude compared to bouton-localized mCherry signals.

- Baseline fluorescence (F_0):
  - Computed as the average pixel intensity of 20 frames, starting from the third frame after recording onset.
    → F0 is average form Frame 3 to 23
    → Qeustion: Is the XLS data already normalized ??

- Fluorescence change (ΔF):
  - Determined by averaging five frames around the peak response after odor onset.

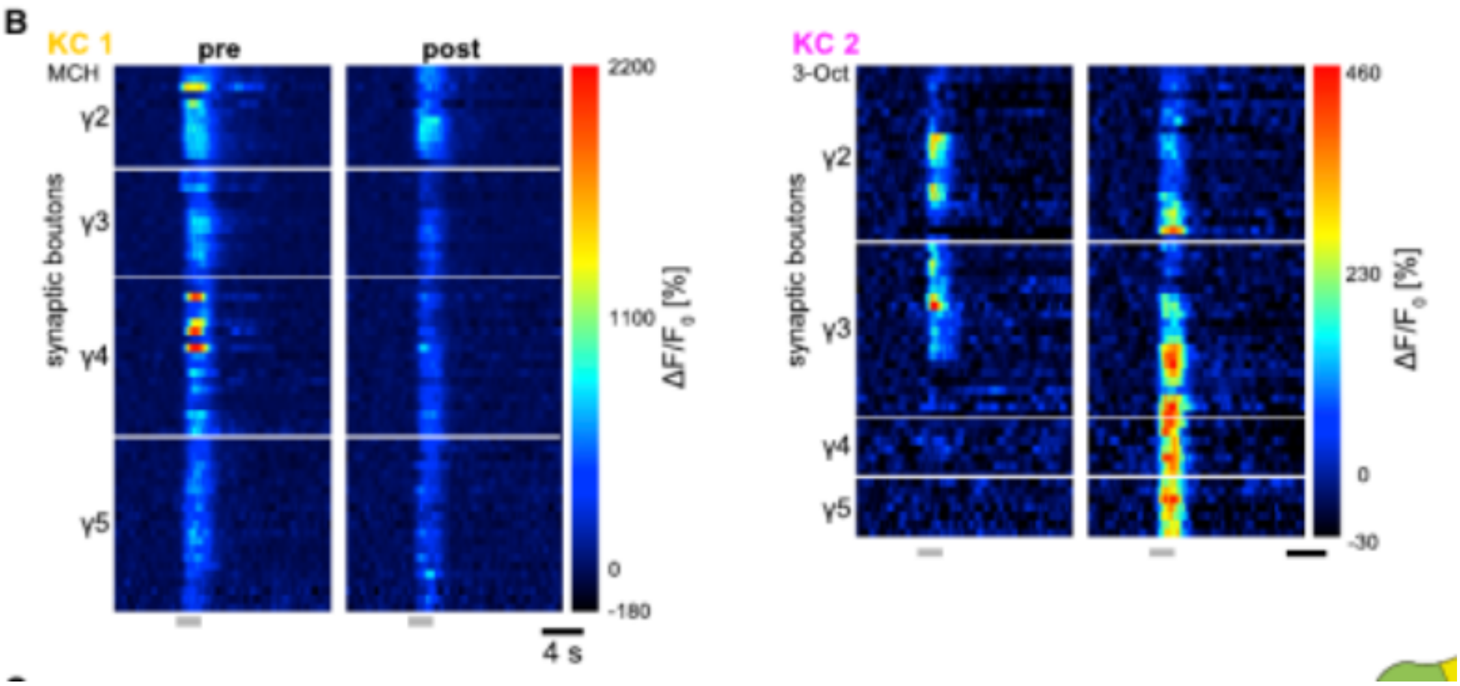
- Noise reduction:
  - A sliding average over 3 frames was applied to smooth the traces.

- Anatomical assignment:
  - Boutons were assigned to γ-lobe compartments using immunohistochemical reference images.

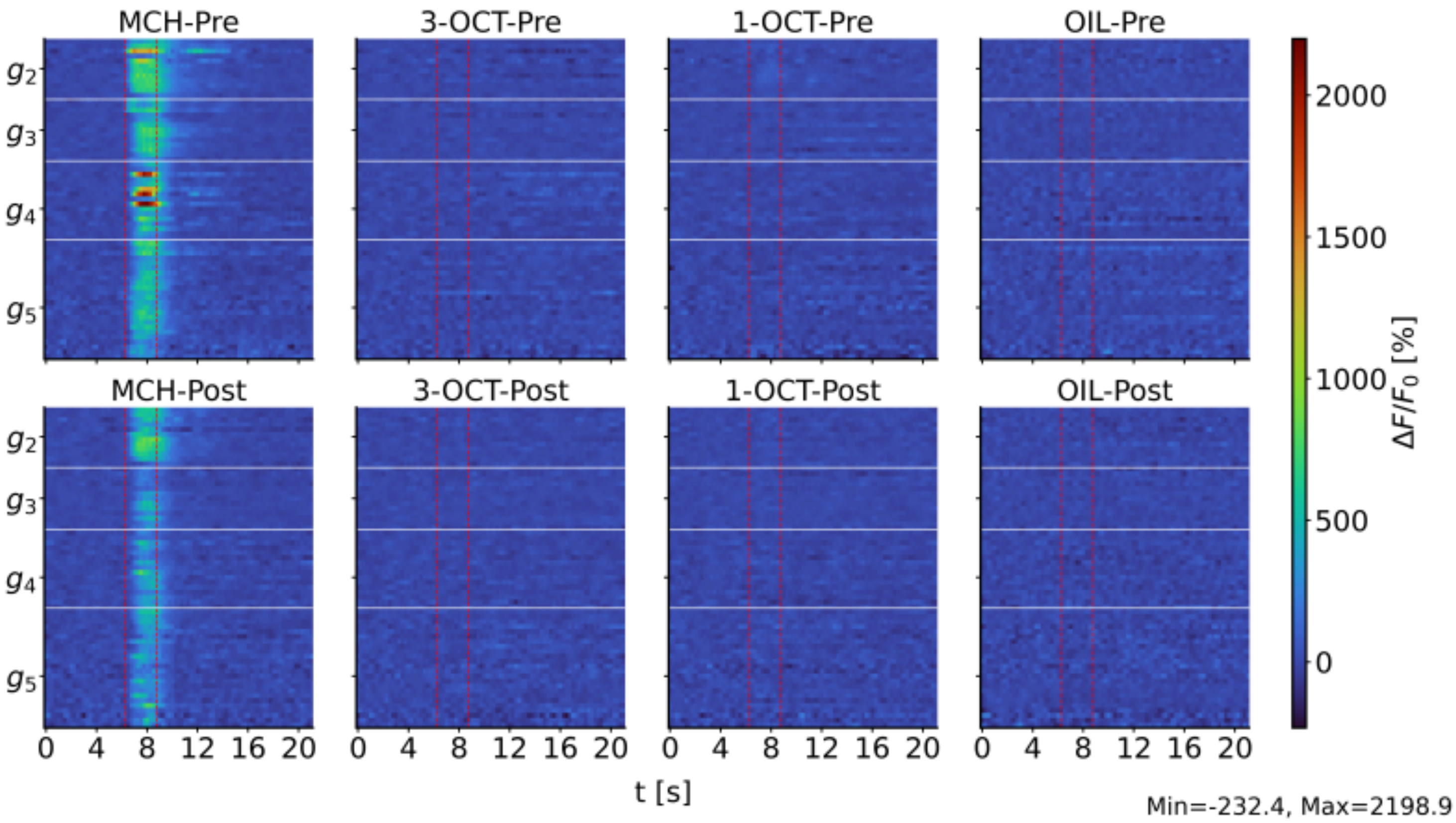
• Data matching

• Tow KC's can be matched form the data set to the paper figures at this point (11.07.25).
  ◇ In ig 5B & Fig 1G 2 KCs are shown: “Kc 1” and "Kc 2". They are the same in both figures and can be matched to specific recordings in the dataset.
  ◇ "KC 1" matches "A_160621_0107_0607_fly1_cell2_pre_post" in the Test group data.
  ◇ "KC 2” matches "A_160621_0107_0607_fly1_cell1_pre_post" in the Test group data.

Fig 5 B (Pre and Post Conditioning state shown)



"A_160621_0107_0607_fly1_cell2_pre_post" Recording AKA “KC 1”



"A_160621_0107_0607_fly1_cell1_pre_post" Recording AKA “KC2”

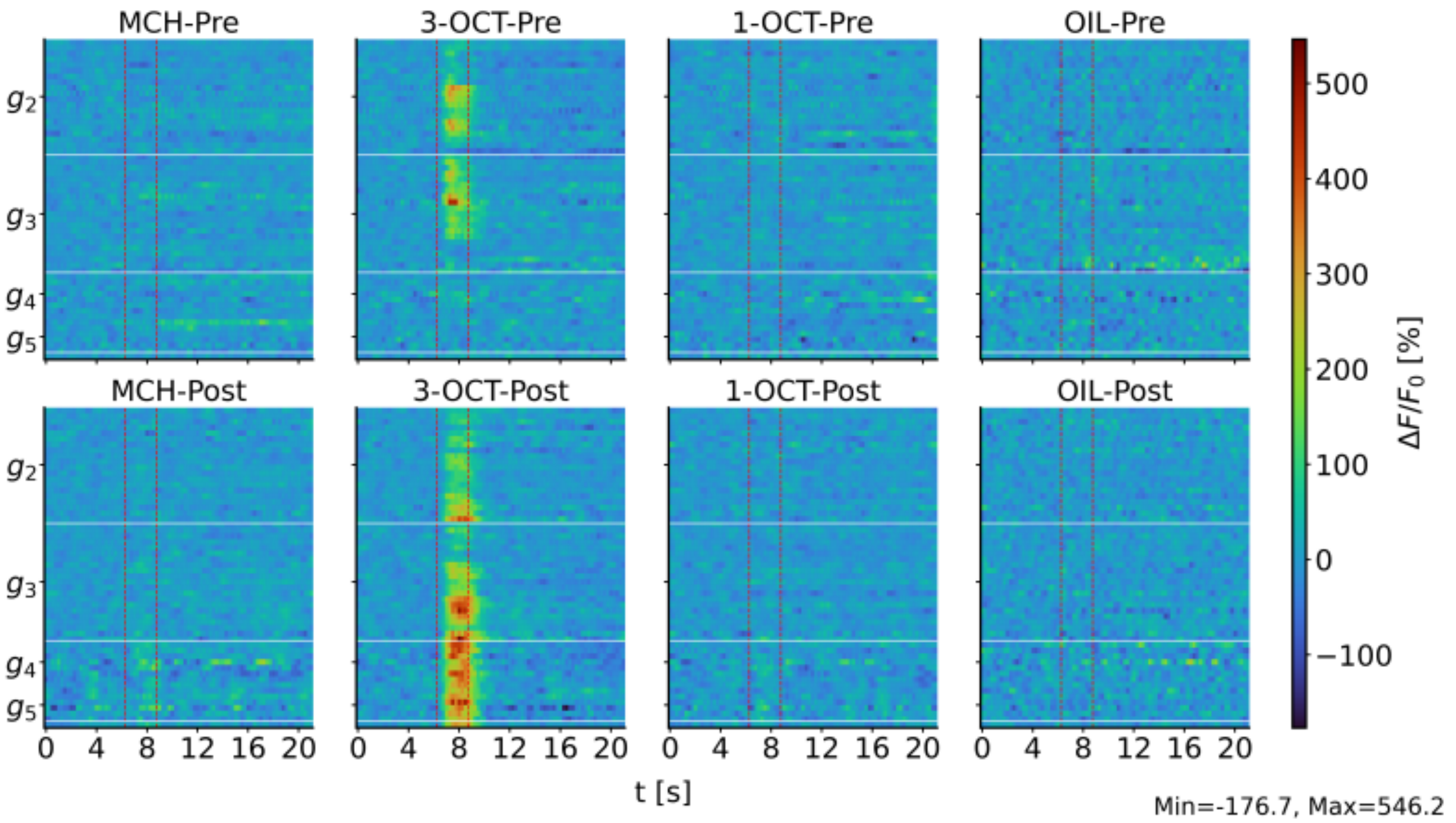
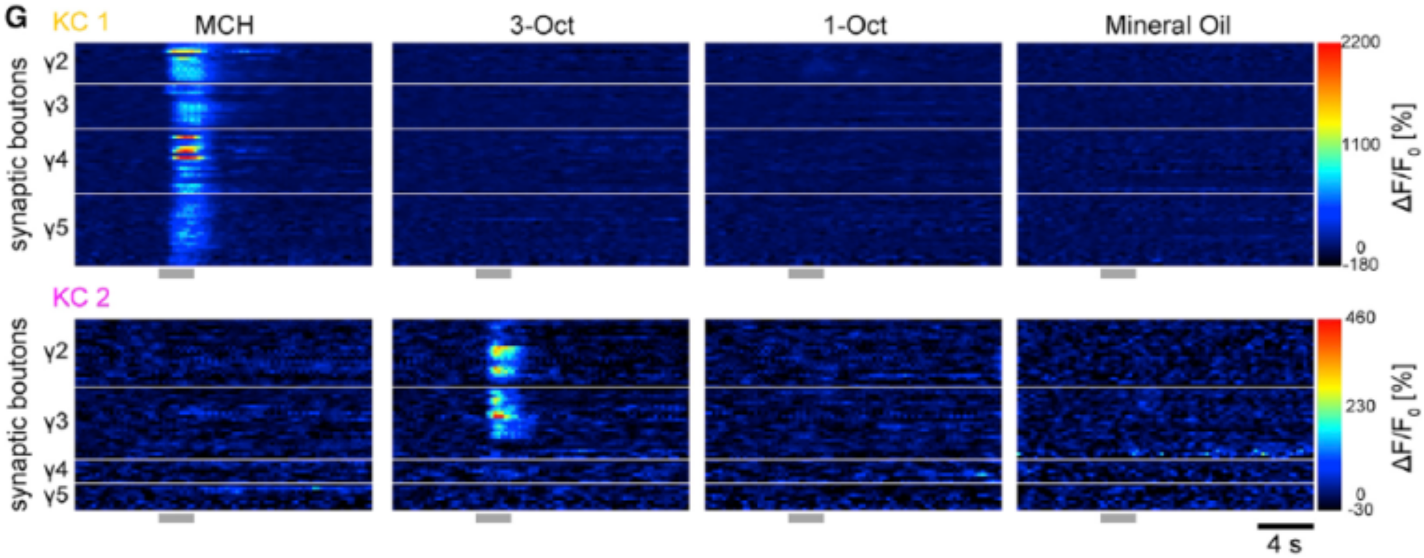


Fig 1 G (Only pre condtioning state is shown)


G
KC 1
MCH
3-Oct
1-Oct
Mineral Oil
synaptic boutons
γ2
γ3
γ4
γ5
2200
1100
0
-180
ΔF/F0 [%]
KC 2
synaptic boutons
γ2
γ3
γ4
γ5
460
230
0
-30
ΔF/F0 [%]
4 s